

IBEH CHIMCHETAM CHRISTABEL

STC Data Analyst Project- Week 3 Report and Dasboards (Health Industry)

Dataset Used

The dataset used for this analysis was sourced from Massachusetts General Hospital and covers a twelve-year period from **2011 to 2022**. It contains records for **974** unique patients and **27,891** hospital encounters.

Business Questions

The following business questions guided the analytical direction of this report:

- What is (are) the highest contributor(s) to the increased operational cost in the hospital?
- How is the increased operational cost affecting other aspects of the Hospital's management?
- Why are patients staying longer than necessary?
- Why are patients leaving inpatient care before the due date?
- Is there a particular demographic associated with longer or shorter stay?
- Is there an inter-relation between increased operational cost and inconsistent patient stay duration?
- Does insurance coverage affect operational cost?
- Is there an inter-relation between insurance coverage and inconsistent patient stay duration?
- What are the causes of unnecessary re-admissions?

Tools Used

All analysis and visualizations for Week 3 were conducted using the cleaned dataset from Week 2 within Excel.

Analysis Performed

Patient Stay & Re-Admissions

I examined patient length of stay (LOS) against likely causal agents, explored the re-admission rate across the full study period and isolated patient segments with the highest frequency of repeated hospitalizations.

- Overall Re-admission Rate: ~60.18% based on data labels.
 - 30-Day Re-admission Rate: 62.51% (encounters occurring within 30 days of a previous discharge).
 - Average Length of Stay (LOS): 0.30 days (~7 hours) across all types, though Inpatient stays average 1.54 days.
 - Departmental Frequency: The Ambulatory department (8,308 cases) and Outpatient (3,357 cases) are the primary sites for re-admissions.
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Patient Demographics

I profiled the patient population using age, gender, race, and mortality indicators, cross-referenced demographic attributes against length of stay to identify groups more susceptible to prolonged or early discharges.

- Average Patient Age: 72.17 years (Range: 26 – 104).
- Mortality Rate: 15.81%, indicating a high-acuity patient population (patients with severe conditions).
- Age Impact: Patients aged 65+ have the highest re-admission rate (64%), representing the hospital's most vulnerable and resource-intensive segment.

Insurance Coverage

I investigated the role of insurance payers in determining patient behaviour (length of stay), and cost burden. Mapped payers against encounter outcomes and stay duration.

- Coverage Status: 68.42% of encounters are insured, leaving 31.58% of encounters as "Uninsured."
- Total insurance coverage paid: \$31,097,507
- Financial Stress: Uninsured encounters show shorter stay durations, correlating with "early self-discharge" patterns driven by cost concerns.
- Payer Impact: Blue Cross Blue Shield patients average the longest stays (2.04 days), while Anthem and Medicare stays are significantly shorter.

Operational Costs

I explored the hospital operational expenditures; identified the highest-cost encounter categories and determined the gap between base costs, claim costs, and payer coverage.

- Total base encounter cost: \$3,240,421
- Total claim cost: \$101,514,376 which is more than 10x higher the base encounter costs
- Total payer coverage: \$31,097,507, which leaves a significant uncovered gap of approximately \$70 million
- Average cost per patient: \$108,334

Dashboard & Visuals

I developed four analytical dashboards to collectively address the business questions:

Patient Stay & Re-Admissions: Visualizes the average length of stay, plots re-admission rates over the study period, and contrasts early discharge patterns against extended-stay cases. Encounter duration trends were highlighted year-by-year, enabling the identification of periods with high turnover.

Patient Demographics: Profiles the hospital's patient population by age group, gender, race, and mortality. This dashboard supports the investigation into whether specific demographic segments (like the 65+ age group) are more prone to longer hospital stays.

Insurance Coverage: Maps the distribution of payers across the 10 active insurance providers, illustrates the proportion of insured versus uninsured encounters, and explores how coverage status relates to patient stay durations.

Operational Costs and Management Insights: Brings together the financial dimensions of the analysis. It examines total claim costs against base encounter costs and payer coverage, quantifying the hospital's uncovered cost burden and identifying high-cost procedure categories like **Electrical cardioversion**.

Key Insights

Re-Admissions Are a Critical Operational Challenge: A re-admission rate of **62.51%** is high and points to issues with post-care follow-up or patient condition severity. This rate directly inflates encounter counts and operational costs.

Gap Between Claims and Payer Coverage: With **\$101.5M** in total claim costs against only **\$31.1M** in payer coverage, the hospital absorbs over **\$70M** in uncovered costs. This gap is a primary driver of operational financial stress.

Elderly Patients Drive Extended Length of Stay: With an average patient age of **72** and a range extending to 104 years, the population is predominantly elderly. Older patients require longer recovery periods and are more prone to re-admission.

Insurance Coverage Gaps Enable Early Discharge: Despite 86% of patients being insured, only **68.42%** of encounters are covered. Uninsured encounters are concentrated among early self-discharges, raising re-admission risk.

High Mortality Rate Signals Patient Acuity: A **15.81%** mortality rate reflects a population with serious conditions demanding more resources per encounter.

Claim Costs Far Exceed Encounter Base Costs: The massive difference between total claim costs (\$101.5M) and base costs (\$3.2M) highlights significant cost accumulation through post-encounter procedures.

Alignment With Business Questions

S/N	Question	Finding
1.	What is the re-admission rate per 30 days?	60.18%
2.	Which procedure incurs the most cost post-insurance?	Electrical cardioversion (Totaling ~\$29.2M in patient/residual costs).
3.	Which department has the highest re-admissions?	Urgent Care (approximately 74% of its encounters result in readmission)
4.	What demographic group stays the longest?	Patients aged 36–50 have the highest average LOS (approximately 29 hours).

5.	Does insurance affect LOS/Readmission?	Yes; Uninsured patients have a significantly higher readmission rate (77.1%).
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